

Intermediate elective

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Set up

Packages

```
# general use and cleaning  
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.2.0      v readr      2.2.0  
v forcats    1.0.1      v stringr    1.6.0  
v ggplot2     4.0.2      v tibble     3.3.1  
v lubridate  1.9.5      v tidyr      1.3.2  
v purrr       1.2.1
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(here)
```

here() starts at /Users/audreyjuneman/Desktop/UCSB/ENVS/ENVS-193DD/github/intermediate-elective

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
library(snakecase)

# wrapping axis labels
library(scales)
```

Attaching package: 'scales'

The following object is masked from 'package:purrr':

discard

The following object is masked from 'package:readr':

col_factor

```
# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)

# colors
library(NatParksPalettes)
```

Data

```
veg <- read_csv(here("data", "veg.csv"))
```

Rows: 27036 Columns: 17

```
-- Column specification -----  
Delimiter: ","  
chr (10): monitors, transect_name, vp_axis, site, transect_side, vp_zone, q...  
dbl (6): x1, transect_length, num_quad, transect_distance, percent_cover, ...  
date (1): date
```

i Use ``spec()`` to retrieve the full column specification for this data.
i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

Rows: 257 Columns: 4

```
-- Column specification -----  
Delimiter: ","  
chr (2): year_pool, transect_name  
dbl (2): year, num_quad
```

i Use ``spec()`` to retrieve the full column specification for this data.
i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
# create a new object from veg  
pools_implicitNA <- veg |>  
  # filter all occurrences where transect_name contains VP  
  filter(str_detect(transect_name, "VP")) |>  
  # group by year, pool, and species  
  group_by(year, transect_name, psoc) |>  
  # calculate mean percent cover  
  summarize(mean_pc = mean(percent_cover, na.rm = TRUE)) |>  
  # ungroup data frame  
  ungroup() |>  
  # create a new column called year_pool from year and transect name  
  unite("year_pool", year, transect_name, remove = TRUE) |>  
  # join with metadata by year_pool  
  left_join(metadata, by = "year_pool")
```

``summarise()`` has regrouped the output.

i Summaries were computed grouped by year, transect_name, and psoc.
i Output is grouped by year and transect_name.
i Use ``summarise(.groups = "drop_last")`` to silence this message.
i Use ``summarise(.by = c(year, transect_name, psoc))`` for per-operation grouping (``?dplyr::dplyr_by``) instead.

```

# create a new object from veg
pools_explicit0 <- veg |>
  # filter all occurrences where transect_name contains VP
  filter(str_detect(transect_name, "VP")) |>
  # complete all possible combinations of year, pool, psoc and distance
  complete(year, transect_name, psoc, transect_distance,
    # fill values of percent cover with 0
    fill = list(percent_cover = 0)
  ) |>
  # group by year, pool, and psoc
  group_by(year, transect_name, psoc) |>
  # calculate total percent cover across all quads
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
  # ungroup the data frame
  ungroup() |>
  # create a new column called year_pool from year and transect name
  unite("year_pool", year, transect_name, remove = TRUE) |>
  # join with metadata data frame
  left_join(metadata, by = "year_pool") |>
  # calculate mean percent cover from total percent cover/number of quadrats
  mutate(mean_pc = sum_pc/num_quad)

```

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- i Use `summarise(.groups = "drop\_last")` to silence this message.
- i Use `summarise(.by = c(year, transect\_name, psoc))` for per-operation grouping (`?dplyr::dplyr_by``) instead.

```

# creating new object from implicit NA data frame
filtered_implicitNA <- pools_implicitNA |>
  # filter to only include
  filter(psoc == "Eryngium_vaseyi")

filtered_explicit0 <- pools_explicit0 |>
  filter(psoc == "Eryngium_vaseyi")

# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

```

Rows: 50 Columns: 15

```
-- Column specification -----  
Delimiter: ","  
chr (15): verbatim_name, taxonRank, scientificName, kingdom, Phylum, Class, ...
```

i Use ``spec()`` to retrieve the full column specification for this data.
i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
# aquatic invertebrates  
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),  
                  sheet = "Aquatic Insects")  
  
# site information  
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),  
                  sheet = "sites")  
  
# water quality information  
water_quality <- read_xlsx(here("data",  
                               "Aquatic Sampling Data-2026-03-10.xlsx"),  
                          sheet = "Water Quality",  
                          na = c("", "n/a", "N/A", "over", "173+"))  
  
# creating new object from sites  
ncos_sites <- sites |>  
  # filter to only include NCOS sites sampled four times a year  
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")  
  
# creating new clean object from taxon_list  
taxon_list_clean <- taxon_list |>  
  # converting taxon name to snake case (no spaces or capital letters,  
  # only underscores)  
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))  
  
# creating new clean object from water quality  
water_quality_clean <- water_quality |>  
  # clean column names  
  clean_names() |>  
  # filtering join: only include sites in ncos_sites  
  semi_join(ncos_sites, by = "site") |>  
  # create new column to join with later data frames  
  unite("date_site", date, site, remove = TRUE) |>  
  # group by date_site column  
  group_by(date_site) |>
```

```

# calculate median pH, salinity, DO
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter
# (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality

```

```

unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal,
                              .fun = "median", .na_rm = TRUE),
       site_full = fct_rev(site_full))

```

`summarise()` has regrouped the output.

- i Summaries were computed grouped by date, site, and taxon.
- i Output is grouped by date and site.
- i Use `summarise(.groups = "drop\_last")` to silence this message.
- i Use `summarise(.by = c(date, site, taxon))` for per-operation grouping
(`?dplyr::dplyr\_by`) instead.